

MODE-S WIND

MODE-S Wind Tactical

**Weather from the sky.
Anywhere. In minutes.**

A complete MODE-S Wind receiving station on a tripod. It listens to the airliners overhead — the wind, temperature and altitude they report themselves — and streams it live into the MODE-S Wind weather picture. No fixed network. No public address. No settings on site.

Set it up, switch it on, and it is feeding.

3	~1 min	150+ NM
steps to deploy	from power-on to live data	farthest aircraft heard

Deploy in three steps

1 — Mount the tripod
Stand it up, raise the mast with the antenna.

2 — Connect antenna & coax
One cable from the antenna to the enclosure.

3 — Power on
The 4G router and the Raspberry Pi 5 — done.

Why MODE-S Wind Tactical

Zero configuration
It joins 4G, opens its own encrypted tunnel to the server and starts feeding by itself. Nothing to type in on site.

Self-healing
A power cut or a restart? About a minute later it is back on line — on its own.

Fills the gaps
Put it where the fixed receivers hear least — low on the approach, behind terrain — and every reply it catches joins the merged weather data.

Goes where you go
Field-tested 8 Nov 2026: from the car boot to live data in 8 minutes, packed in 5, about 10 W from a portable battery.

Proven on live data
Already feeding as MODES-RX-02: 60–95 messages a second in the daytime, about 5 % of all merged replies heard by it alone.

Light on data
About 80–100 MB of mobile data a day — an ordinary 4G subscription is enough.

INSIDE

1090 MHz antenna (OPA-5 series, 5 dBi) · low-loss H-155 coax · AirSpy R2 receiver · Raspberry Pi 5 · mobile 4G router · tripod with telescopic mast

See it live:
mode-s-wind.oh2gax.org

MODE-S Wind — Otso Laakso (OH2GAX), November 2026. Figures from the live system, 7 Nov 2026, and the field test of 8 Nov 2026. A research system: not for flight operations.